Problem C
Integer After Removing Similar Binaries

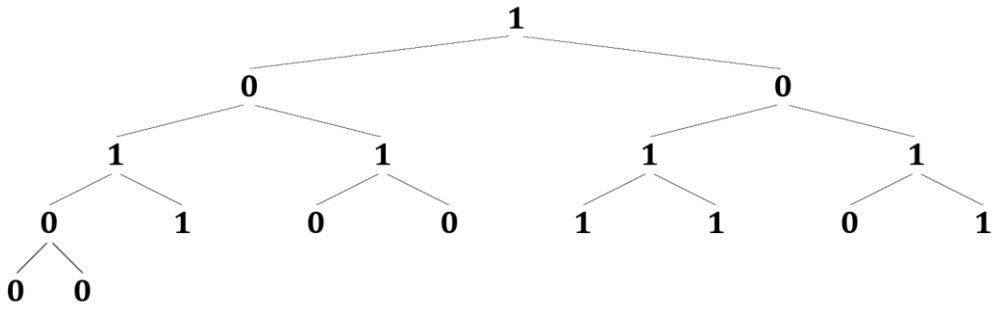
Time limit: 1 second

You are given a positive integer n. Write it in binary, most significant bit first, without leading zeros. Place its bits in a complete binary tree, filling each level from left to right.

A subtree consists of a node and all its descendants. Remove an entire subtree, including its root, if it contains at least three nodes and all its bits are the same.

Visit the tree in preorder: root, left subtree, then right subtree. Check each node before its children. If its subtree is removed, skip all its nodes. Otherwise, keep its root and continue. Do not check a node again after pruning its descendants.

The diagram shows the tree for $n = 81204$.



Removing the three-node all-zero subtree on the left and the three-node all-one subtree on the right leaves the preorder sequence $10111000101_2 = 1477$.

Input

A single positive integer n $(1 \leq n \leq 10^9)$

Output

Print the decimal value of the remaining preorder bit sequence. If no nodes remain, print 0.

Sample Input 1	Sample Output 1
81204	1477
Sample Input 2	**Sample Output 2**
128	1
Sample Input 3	**Sample Output 3**
1028	17
Sample Input 4	**Sample Output 4**
1031	139